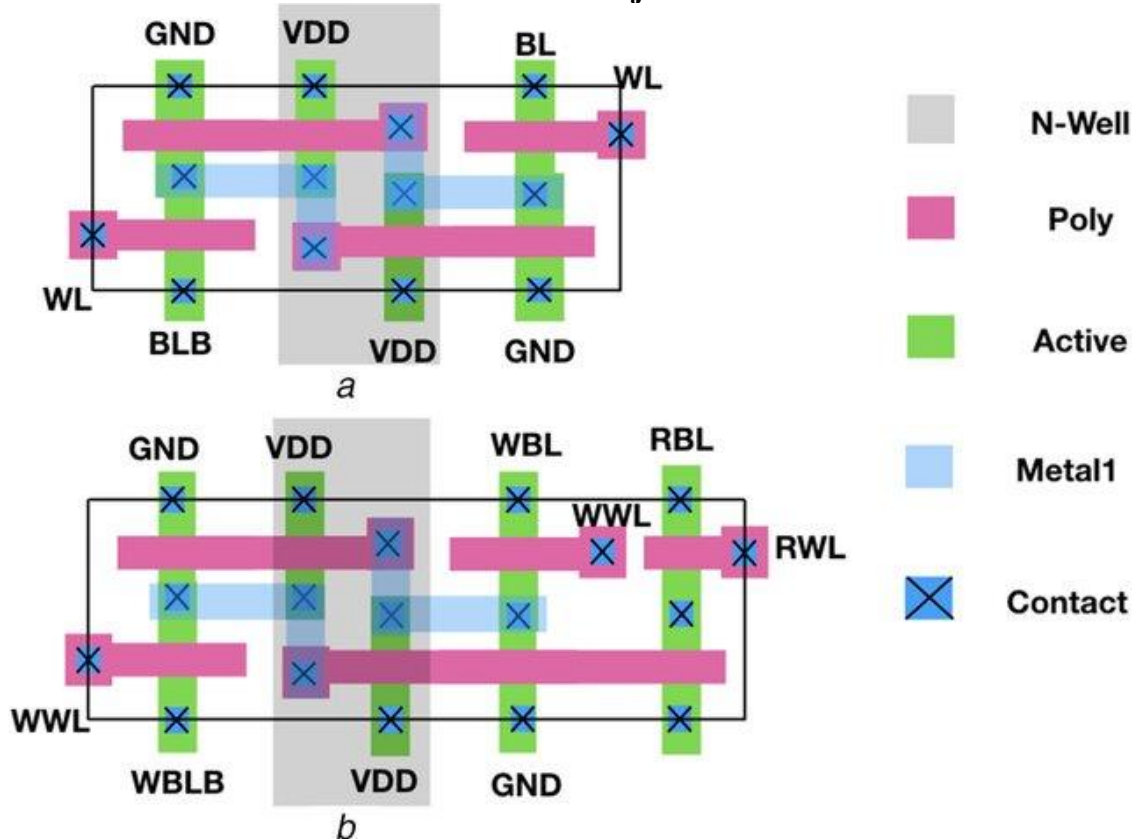




Course Project



You are given the symbolic layout shown in the figure. It is required to draw the extracted schematic and then redraw its layout using any suitable software. The layout should pass the LVS and DRC tests.

You can use any suitable software for schematic and layout:

- 1- Cadence
- 2- Microwind <https://www.microwind.net/>
- 3- Electric <http://www.staticfreesoft.com>

(It is recommended to begin using Electric if you have no background on any simulation tools)

Deliverables:

- A documentation for the project reporting the schematic, layout drawn, DRC results, LVS results etc.
- A .zip file containing all your project files (schematic and layout).

The report should be uploaded to the LMS before the **8th of May at 11:59 PM**.

General Instructions:

- Use any available process design kit (PDK)
- Use any sizing you see appropriate for the circuits.

Evaluation Criteria

1	Layout	30%
2	Schematics	30%
3	DRC	20%
4	LVS	20%

Bonus (5%)

Extract the parasitics of the layout and do post-layout simulations showing the difference between figures a and b.

Academic Misconduct

The University defines Academic Misconduct as ‘any case of deliberate, premeditated cheating, collusion, plagiarism or falsification of information, in an attempt to deceive and gain an unfair advantage in assessment’. This includes attempting to gain marks as part of a team without making a contribution. The department takes Academic Misconduct very seriously and any suspected cases will be investigated through the University’s standard policy. If you are found guilty, you may be expelled from the University with no award.

It is your responsibility to ensure that you understand what constitutes Academic Misconduct and to ensure that you do not break the rules. If you are unclear about what is required, please ask.

Guide to Marking Criteria (Please read the accompanying marking scheme)

Course Code:		Course Name:		Assignment No.		Date:		
Student Name:					Student ID:			

	Mastery ≥90%				Accomplished ≥75 & <90%				Adequate ≥60 & <75%				Inadequate <60%			
	100	96	93	90	89	84	79	75	74	69	64	60	59	40	20	0

Layout (30%)	Layout complete and no wrong connections or layers.	Few wrong connections or layers.	Many wrong connections and layers.	All connections and layers are wrong.
Schematics (30%)	Circuit diagram is correct, and all the connections are correct.	Few wrong connections and transistors.	Many wrong connections and transistors.	All connections and transistors are wrong.
DRC (20%)	Zero DRC errors.	Few DRC errors.	Many DRC errors.	DRC didn't run or all errors.
LVS (20%)	Zero LVS errors.	Few LVS errors.	Many LVS errors.	LVS didn't run or all errors.

1 st marker Total		1 st marker Signature		Agreed	
2 nd Marker Total		2 nd marker Signature		Mark	

General Comments: 	UEL Grading System	Agreed Mark Range	ASU Grading Scale	
	% equivalent at UEL		% at ASU	Grade
	95% and higher		97% and higher	A+
	82% to less than 95%		93% to less than 97%	A
	70% to less than 82%		89% to less than 93%	A-
	66% to less than 70%		84% to less than 89%	B+
	63% to less than 66%		80% to less than 84%	B
	60% to less than 63%		76% to less than 80%	B-
	56% to less than 60%		73% to less than 76%	C+
	53% to less than 56%		70% to less than 73%	C
	50% to less than 53%		67% to less than 70%	C-
	45% to less than 50%		64% to less than 67%	D+
	40% to less than 45%		60% to less than 64%	D
Less than 40%	Less than 60%	F		